

Week 1 Report

Algorithmic Trading — Summer of Science

Arush Anand

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Chapter 1 Summary: *Quantitative Trading* by E. P. Chan

What Is Quantitative Trading?

Chan (2021) says that Quantitative Trading is the trading of securities based strictly on the buy/sell decisions of computer algorithms. It is not quite a technical analysis as only metrics that are quantifiable (as opposed to them being qualitative) are used in Quantitative Trading.

Who Can Become a Quantitative Trader?

In his own words, "The ideal independent quantitative trader is therefore someone who has some prior experience with finance or computer programming, who has enough savings to withstand the inevitable losses and periods without income, and whose emotion has found the right balance between fear and greed." This ideal trader might not have the academic qualifications of the stereotypical algorithmic trader, as that's not a prerequisite or even that big of an edge over the layman.

The Business Case for Quantitative Trading

Chan frames a quantitative trading operation as a small business and then identifies three ways in which it differs favourably from most other ventures.

Scalability

Increasing the size of a quantitative portfolio often requires nothing more than adjusting a leverage parameter in code. No venture capital, no bank negotiation. Proprietary trading firms can offer intraday leverage of up to $\times 40$ on equity, while futures and currency accounts routinely provide $\times 10$ or more from standard brokerages. Chan warns, however, that leverage is a double-edged sword and should never be chased for its own sake.

Demand on Time

Automation is intrinsic to algorithmic trading. In Chan's own workflow, day-to-day operations (data downloads, signal generation, order transmission) initially consumed roughly two and a half hours around the market open and close; over time these were fully automated until active intervention fell to zero. Research and strategy development can be scheduled flexibly. The largest psychological challenge, Chan admits, is resisting the urge to manually override positions during intraday drawdowns.

No Marketing Required

In most businesses revenue depends on persuading customers. In trading, the counterparty's only concern is price. As long as a trader is not managing outside money, the

product (the strategy) markets itself through its P&L. This freedom from salesmanship is, for many practitioners, the greatest appeal of the profession.

Key takeaways from Chapter 1:

- Quantitative trading is a form of systematic, algorithm-driven trading.
- In a search for complex strategies, it is usually the simpler ones which would benefit us more.
- The business advantages are scalability, time efficiency through automation, and zero marketing overhead.

Kaggle Pandas Certificate

As part of the Week 1 curriculum I completed the **Pandas** course on Kaggle, which covers the core data-manipulation library used throughout quantitative research in Python. The certificate of completion is reproduced below.



Figure 1: Kaggle Pandas course certificate — Arush Anand.

References

Ernest P. Chan. *Quantitative Trading: How to Build Your Own Algorithmic Trading Business*. John Wiley & Sons, Hoboken, New Jersey, 2nd edition, 2021. ISBN 9781119800064.